



## **PROFESSIONAL SUMMARY**

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- Experienced Data Engineer with 7+ years of expertise in designing, developing, and optimizing scalable data pipelines, ETL workflows, and cloud-based solutions across Hadoop, AWS and Azure.
- Strong cloud expertise in AWS - S3, Redshift, EMR, Glue, Lambda, Athena, DynamoDB, Kinesis, RDS and Azure - Data Factory, Synapse Analytics, Databricks, ADLS, Blob Storage, Cosmos DB, Power BI for big data processing and analytics.
- Strong hands-on experience in the Hadoop ecosystem, including HDFS, MapReduce, YARN, Apache Hive, Sqoop, Oozie, and Spark, ensuring efficient large-scale data processing.
- Expertise in big data technologies such as Apache Spark, Snowflake, Delta Lake, Hive, and Kafka, ensuring high-performance data transformation and real-time processing.
- Hands-on experience with Python, SQL, and PySpark, optimizing data processing, automation, and machine learning workflows.
- Developed real-time data streaming solutions using AWS Kinesis on Ericsson and Azure Event Hubs on Walmart, enabling low-latency insights for business intelligence and analytics.
- Developed scalable batch and real-time processing solutions using Hadoop MapReduce, Apache Spark, and Kafka, improving data transformation efficiency in financial applications.
- Machine Learning & NLP experience using AWS SageMaker, Azure ML, TensorFlow, Scikit-learn, and NLP tools like SpaCy, NLTK for predictive modeling, anomaly detection, and sentiment analysis.
- Data warehouse expertise in AWS Redshift on Ericsson and Azure Synapse on Walmart, optimizing data storage, indexing, and query performance for large-scale datasets.
- ETL & Workflow Orchestration using AWS Glue, Step Functions, Apache Airflow, and Azure Data Factory, ensuring seamless automation and scheduling.
- DevOps & Infrastructure Automation experience with Terraform, AWS CloudFormation, and Azure DevOps, streamlining cloud deployments and CI/CD pipelines.
- Data Security & Compliance expertise in IAM, RBAC, KMS encryption, GDPR, and HIPAA, ensuring secure and compliant data operations across cloud platforms.
- Designed and implemented business intelligence solutions using Power BI and AWS Quick Sight, providing real-time insights into financial, telecom, and retail data.
- Proven ability to work in fast-paced, Agile environments, collaborating with cross-functional teams to drive data-driven decision-making and operational efficiency.
- Expert in large-scale data migrations, having successfully transitioned on-premises legacy systems to cloud-native architectures in AWS on Chase and Azure on Walmart, improving scalability and cost efficiency.
- Optimized Spark-based ETL jobs on AWS EMR and Azure Databricks, improving performance, cost-efficiency, and processing times for high-volume datasets.
- Automated data workflows using Apache Oozie and Airflow, reducing manual ETL effort by 50% and improving job scheduling efficiency.
- Proficient in DevOps for Hadoop environments, implementing CI/CD pipelines with Terraform, Jenkins, and AWS CloudFormation to streamline deployments.
- Developed Hadoop-based ETL pipelines using Apache Hive, Sqoop, and Oozie, optimizing data ingestion and processing for large-scale financial datasets.
- Built and managed a Hadoop-based data lake integrating HDFS, Hive, and Spark, improving data storage, retrieval, and reporting efficiency.
- Optimized Hadoop MapReduce jobs, reducing batch processing time by 40%, enhancing fraud detection and risk analytics performance.
- Integrated Kafka with Hadoop to develop real-time data streaming pipelines, ensuring low-latency transaction monitoring.

- Leveraged YARN for Hadoop resource management, optimizing distributed computing performance across large datasets
- Developed automated anomaly detection systems using machine learning models in AWS SageMaker and Azure ML, reducing fraudulent transactions and improving data integrity.
- Designed and implemented data governance frameworks, leveraging AWS Lake Formation and Azure Purview, ensuring role-based access control, encryption, and compliance with GDPR and HIPAA.
- Strong problem-solving skills, with experience in troubleshooting and performance tuning of data pipelines, query optimization in Redshift - AWS and Synapse Analytics -Azure, and reducing latency in real-time streaming solutions.

## **TECHNICAL SKILLS**

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|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming Languages</b>                  | Python, SQL, PySpark, R, Shell Scripting, Bash                                                                                                                                                                        |
| <b>Databases</b>                              | SQL Server, PostgreSQL, MySQL, Oracle, DynamoDB, Cosmos DB                                                                                                                                                            |
| <b>Cloud Infrastructure</b>                   | <b>AWS Cloud</b> – S3, Redshift, EMR, Glue, Lambda, Athena, DynamoDB, Kinesis, RDS, EC2, IAM, CloudFormation<br><b>Azure Cloud</b> – Data Factory, Synapse, Databricks, Blob Storage, Cosmos DB, Event Hubs, Power BI |
| <b>Big Data Tools</b>                         | Apache Spark, Snowflake, Delta Lake, Hive, Kafka, HDFS, Hadoop( HDFS, Yarn, Mapreduce)                                                                                                                                |
| <b>ETL &amp; Data Pipelines:</b>              | AWS Glue, Azure Data Factory (ADF), Apache Airflow, dbt, Oozie, Hadoop-based ETL workflows                                                                                                                            |
| <b>Data Warehousing:</b>                      | SQL Server, PostgreSQL, MySQL, Amazon Redshift, Azure Synapse Analytics, Snowflake                                                                                                                                    |
| <b>Data Modeling &amp; Processing:</b>        | Star Schema, Snowflake Schema, Data Lakes, Fact-Dimension Modeling.                                                                                                                                                   |
| <b>Business Intelligence &amp; Reporting:</b> | Power BI, Tableau, AWS QuickSight                                                                                                                                                                                     |
| <b>Workflow Orchestration:</b>                | Apache Airflow, Oozie, Azure Data Factory, AWS Step Functions                                                                                                                                                         |
| <b>DevOps &amp; CI/CD:</b>                    | Docker, Kubernetes, Terraform, AWS CloudFormation, Jenkins, Git                                                                                                                                                       |
| <b>Security &amp; Compliance:</b>             | IAM, RBAC, KMS Encryption, GDPR, HIPAA, Data Masking                                                                                                                                                                  |
| <b>Infrastructure as Code (IaC):</b>          | Terraform, AWS CloudFormation                                                                                                                                                                                         |
| <b>Version Control &amp; Collaboration:</b>   | Git, GitHub, Bitbucket                                                                                                                                                                                                |
| <b>Machine Learning &amp; AI Frameworks:</b>  | AWS SageMaker, Azure ML, TensorFlow, Scikit-learn, Keras, PyTorch, NLP (SpaCy, NLTK), Pandas, NumPy                                                                                                                   |

## **PROFESSIONAL EXPERIENCE:**

**Chase Bank, New York, NY**

**Dec2023–Present**

**Data Engineer**

- Designed and implemented scalable data pipelines on AWS using Glue, Lambda, Step Functions, and S3, ensuring efficient ETL workflows for structured and semi-structured data.
- Developed and optimized AWS Redshift-based data warehouses, leveraging Spectrum and Athena for interactive queries and real-time insights.
- Built and managed data lake architectures using Amazon S3, incorporating partitioning and compression strategies to enhance performance and cost efficiency.
- Integrated Kinesis Data Streams and Firehose for real-time data ingestion and processing, enabling low-latency analytics on financial transactions.

- Developed serverless data processing solutions using AWS Lambda and Glue PySpark scripts, automating data transformations and aggregation at scale.
- Developed a custom orchestrator using Apache Airflow & AWS Step Functions to schedule, monitor, and manage ETL dependencies.
- Designed a job scheduling framework in AWS Glue that enabled automatic retries, parallel execution, and optimized resource usage.
- Implemented dependency-aware workflows in Airflow DAGs, ensuring efficient execution of upstream and downstream tasks.
- Developed PowerShell scripts to automate ETL deployments, database migrations, and file system management in AWS S3 and Azure Blob Storage.
- Optimized batch jobs for data movement between on-prem SQL Server and AWS Redshift, improving job execution time by 30%.
- Used PowerShell scripts for automated log cleanup, file archiving, and data validation in S3, reducing manual intervention.
- Developed batch jobs to process and transform structured/unstructured data files CSV, JSON before ingestion into the data warehouse.
- Leveraged YARN for Hadoop resource management, optimizing distributed computing performance across large datasets.
- Implemented data security best practices in Hadoop clusters using Kerberos authentication, Apache Ranger, and data encryption, ensuring compliance with GDPR & PCI-DSS..
- Integrated Hadoop with AWS services (EMR, S3, Glue, Redshift), enabling hybrid cloud data processing architectures.
- Designed and tuned SQL & HiveQL queries for Hadoop-based data lakes, reducing query execution time by 30% for real-time reporting.
- Automated Hadoop job scheduling and monitoring using Apache Oozie, improving ETL workflow efficiency and failure recovery.
- Developed predictive analytics models using Hadoop-based Spark MLlib, enhancing fraud detection and credit risk analysis. Ensured data quality, security, and compliance by implementing AWS Lake Formation, IAM roles, encryption (KMS), and fine-grained access controls.
- Orchestrated complex ETL workflows using Apache Airflow and AWS Step Functions, optimizing dependencies and job execution timelines.
- Migrated legacy on-premises SQL databases to AWS RDS (PostgreSQL & MySQL) and DynamoDB, improving scalability and operational efficiency.
- Developed and optimized SQL queries for complex data transformations, indexing, partitioning, and query tuning across AWS Redshift, RDS, and Snowflake, ensuring efficient data retrieval and processing.
- Built predictive models for risk analysis and fraud detection using Python, Scikit-learn, and AWS SageMaker, leveraging historical financial transaction data to identify anomalies and prevent fraudulent activities.
- Designed NLP-powered data extraction pipelines utilizing AWS Textract, Comprehend, and Python-based NLP libraries (SpaCy, NLTK) to automate document processing and entity recognition in financial reports.
- Implemented advanced data validation techniques using PySpark, SQL, and AWS Glue, ensuring data accuracy, consistency, and compliance with Chase's financial regulations.
- Automated data ingestion from external financial APIs and third-party data sources using AWS Lambda, Step Functions, and API Gateway, enabling real-time integration with internal data platforms.
- Leveraged Azure Data Factory (ADF) and Synapse Analytics for cross-cloud data integration, enabling seamless data movement between AWS and Azure environments.
- Optimized Spark-based ETL jobs on AWS Glue & EMR, tuning resource configurations for cost-effective processing of high-volume financial datasets.
- Monitored and troubleshoot data pipeline performance using AWS CloudWatch, CloudTrail, and Datadog, ensuring high availability and reliability.

- Implemented data governance best practices, enforcing data lineage tracking, metadata management, and audit logging across cloud platforms.
- Collaborated with cross-functional teams to build CI/CD pipelines using Terraform, CloudFormation, and Git, ensuring automated deployment of data solutions.

**Environment:** AWS: S3, Redshift, EMR, Glue, Lambda, Athena, Kinesis, DynamoDB, RDS, IAM, CloudFormation, Step Functions, Lake Formation, CloudWatch, CloudTrail | Azure: Azure Data Factory (ADF), Synapse Analytics, Databricks, Blob Storage | Big Data & ETL: Apache Spark, PySpark, Snowflake, Delta Lake, Hive, Apache Airflow, Oozie | Programming & Scripting: Python, SQL, R, Shell Scripting, Bash | Machine Learning & NLP: AWS SageMaker, Scikit-learn, TensorFlow, Keras, AWS Textract, AWS Comprehend, NLTK, SpaCy | Data Streaming: Apache Kafka, AWS Kinesis, Azure Event Hubs | Databases: SQL Server, PostgreSQL, MySQL, DynamoDB, Cassandra, MongoDB | DevOps & CI/CD: Terraform, AWS CloudFormation, Docker, Kubernetes, Jenkins, Git | Hadoop: Hadoop-based ETL workflows | PowerShell & Batch Scripting | Data Orchestration

## **Walmart, Bentonville, AR** **Engineer**

**Jan 2022 – Nov2023 Data**

- Designed and implemented scalable ETL pipelines using Azure Data Factory (ADF), Synapse Analytics, and Databricks, processing large volumes of retail transaction data.
- Developed and optimized data lake architectures using Azure Data Lake Storage (ADLS), ensuring efficient storage, partitioning, and cost-effective data retrieval for analytics.
- Built real-time data streaming solutions using Azure Event Hubs and Stream Analytics, enabling low-latency insights into customer behavior, inventory management, and supply chain optimization.
- Implemented big data processing workflows using Azure Databricks (PySpark, Scala), optimizing data transformations, aggregations, and machine learning model execution.
- Designed and implemented idempotent ETL pipelines in Azure Data Factory (ADF) to ensure that interrupted, incomplete, or failed processes could be re-run without data duplication or inconsistency.
- Utilized parameterized pipelines and checkpointing mechanisms to enable seamless reruns of failed jobs in ADF.
- Automated error logging and notifications in ADF using Azure Monitor & Log Analytics, ensuring real-time failure tracking.
- Integrated Snowpipe with Azure Blob Storage, enabling automated real-time ingestion of streaming data into Snowflake.
- Optimized Snowpipe performance by configuring auto-ingest with external stage and partitioning strategies.
- Developed RESTful APIs in Azure Data Factory (ADF) to enable real-time data retrieval and updates from external applications.
- Integrated Snowflake with API Services by developing external functions to pull and push data between Snowflake and third-party APIs.
- Designed and optimized data warehouses using Azure Synapse Analytics, ensuring high-performance querying and real-time analytics for Walmart's retail operations.
- Automated data ingestion from multiple sources (APIs, IoT, external vendors) using Azure Logic Apps and ADF, ensuring seamless data movement and transformation.
- Developed high-performance SQL queries for Azure Synapse Analytics and Databricks, optimizing query execution for large-scale retail datasets and improving reporting efficiency.
- Built predictive models for demand forecasting, customer segmentation, and inventory optimization using Azure Machine Learning (Azure ML) and Databricks ML.
- Implemented Natural Language Processing (NLP) solutions using Azure Cognitive Services and Python (SpaCy, NLTK) to analyze customer reviews, sentiment, and feedback trends.
- Developed Power BI dashboards integrated with Azure Synapse and ADLS, providing Walmart's business teams with real-time insights into sales performance and customer engagement trends.
- Designed and implemented Hadoop-based ETL pipelines using Apache Hive and Spark, optimizing data ingestion and transformation for large-scale retail analytics.

- Developed a Hadoop-powered data lake using HDFS and Delta Lake, enabling efficient storage and retrieval of sales, inventory, and customer transaction data.
- Integrated Apache Kafka with Hadoop for real-time event processing, ensuring low-latency insights into inventory movement and demand forecasting.
- Optimized SQL and HiveQL queries on Hadoop-based data lakes, reducing data processing time by 30%, improving reporting and analytics efficiency.
- Automated batch processing workflows using Apache Oozie and Airflow, streamlining data pipeline orchestration and reducing operational overhead.
- Migrated on-premises data warehouses to Azure Synapse, leveraging PolyBase and Azure Data Factory, improving scalability and reducing operational costs.
- Optimized Azure Databricks jobs by tuning cluster configurations, caching strategies, and parallel execution, improving processing times for high-volume retail datasets.
- Implemented CI/CD pipelines for data pipeline deployment and infrastructure automation using Azure DevOps, Terraform, and Git, ensuring seamless integration across teams.
- Developed automated monitoring and alerting systems using Azure Monitor, Log Analytics, and Application Insights, ensuring real-time issue detection and resolution.
- Collaborated with supply chain, finance, and customer analytics teams, translating business requirements into scalable data solutions that drive operational efficiency and decision-making.

**Environment:** Azure: Azure Data Factory (ADF), Synapse Analytics, Azure Databricks, Azure Data Lake Storage (ADLS), Blob Storage, Cosmos DB, Azure ML, Azure Cognitive Services, Event Hubs, Power BI, Logic Apps, Purview | Big Data & ETL: Apache Spark, PySpark, Snowflake, Delta Lake, Hive, Apache Airflow, dbt | Programming & Scripting: Python, SQL, R, Shell Scripting, Bash | Machine Learning & NLP: Azure ML, Databricks ML, TensorFlow, Scikit-learn, Keras, Azure Cognitive Services (Text Analytics, Speech-to-Text), SpaCy, NLTK | Streaming & Real-Time Processing: Azure Event Hubs, Kafka | Databases: Azure Synapse Analytics, SQL Server, PostgreSQL, MySQL, Cosmos DB | DevOps & Security: Azure DevOps, Terraform, Docker, Kubernetes, Git, IAM, RBAC, GDPR | Monitoring & Logging: Azure Monitor, Log Analytics, Application Insights | ADF | Snowflake | Snowpipe | ETL Design

## Ericsson, Bengaluru, India Data Engineer

July 2020 – Aug 2021

- Designed and developed scalable ETL pipelines using AWS Glue, Lambda, Step Functions, and S3, optimizing data processing for telecom network analytics.
- Built and managed data lake architectures on Amazon S3, implementing partitioning, compression, and lifecycle policies to enhance performance and cost efficiency.
- Developed real-time data streaming solutions using AWS Kinesis (Data Streams & Firehose), enabling low-latency processing for telecom network monitoring and predictive maintenance.
- Developed and maintained Java-based microservices for real-time data processing within the Ericsson data streaming platform, contributing to enhanced system performance.
- Implemented Java components for data transformation and enrichment within the AWS Kinesis data pipelines, ensuring data integrity and accuracy.
- Developed multithreaded Java applications for processing high-volume network traffic data, improving the efficiency of real-time analytics.
- Created and maintained JavaServer Pages (JSPs) for dynamic web interfaces within Ericsson's internal data management tools
- Implemented big data processing workflows with Apache Spark (PySpark) on AWS EMR, optimizing data transformations, aggregations, and feature engineering.
- Designed and optimized data warehouses using AWS Redshift, leveraging Spectrum and Athena for fast and cost-effective querying of large telecom datasets.

- Automated data ingestion and transformation from multiple sources (IoT devices, network logs, customer interactions) using AWS Glue, Lambda, and Step Functions.
- Developed Python- and SQL-based data validation frameworks, ensuring data integrity, accuracy, and compliance with telecom industry regulations.
- Built machine learning models for network performance optimization and anomaly detection using AWS SageMaker, Scikit-learn, and TensorFlow, reducing downtime and improving fault detection.
- Established data security and governance best practices using AWS IAM, KMS encryption, and Lake Formation, ensuring fine-grained access control and compliance.
- Implemented CI/CD pipelines for data engineering solutions using AWS CodePipeline, Terraform, and CloudFormation, automating infrastructure deployment and scaling.
- Monitored and optimized ETL and data workflows using AWS CloudWatch, CloudTrail, and Datadog, ensuring high availability and proactive performance tuning.
- Developed business intelligence dashboards using AWS QuickSight, providing real-time insights into network performance, traffic patterns, and customer experience analytics.
- Led cross-functional collaboration with data scientists, network engineers, and business teams, translating complex telecom data into actionable insights.
- Migrated on-premises data platforms to AWS cloud-native solutions, improving scalability, system reliability, and cost optimization.

**Environment:** AWS: S3, Redshift, EMR, Glue, Lambda, Athena, Kinesis (Data Streams & Firehose), RDS, IAM, CloudFormation, Step Functions, SageMaker, QuickSight, Lake Formation | Big Data & ETL: Apache Spark, PySpark, Hive, Snowflake, Delta Lake, Apache Airflow | Programming & Scripting: Python, SQL, R, Shell Scripting, Bash | Machine Learning & AI: AWS SageMaker, Scikit-learn, TensorFlow, Keras, PySpark MLlib | Streaming & Real-Time Processing: AWS Kinesis, Kafka | Databases: Redshift, SQL Server, PostgreSQL, MySQL, DynamoDB | Java, Java Server page | DevOps & Security: AWS CodePipeline, Terraform, CloudFormation, IAM, RBAC, KMS Encryption | Monitoring & Logging: AWS CloudWatch, CloudTrail, Datadog

## **Axis Bank, Mumbai, India**

**May 2017- July 2020 Data**

### **Analyst**

- Analyzed large-scale financial datasets from SQL Server, Oracle, and PostgreSQL, identifying key trends, anomalies, and business insights to support risk and compliance teams.
- Designed and developed interactive dashboards and reports using Power BI and Tableau, providing real-time insights into customer transactions, loan performance, and fraud detection.
- Wrote complex SQL queries, stored procedures, and views to extract, clean, and aggregate structured data from core banking systems, improving data accuracy and reporting efficiency.
- Conducted predictive analytics and statistical modeling using Python, Pandas, NumPy, Scikit-learn to enhance customer segmentation and improve targeted marketing campaigns.
- Implemented data quality and validation frameworks, ensuring data consistency, integrity, and compliance with RBI- Reserve Bank of India and other regulatory guidelines.
- Automated monthly and quarterly financial reporting processes, reducing manual effort and improving report turnaround time by 40%.
- Assisted in migrating legacy Excel-based reporting to cloud-based BI solutions like Power BI, SQL Server Reporting Services, enabling real-time access to financial data.
- Utilized Java Spring Boot to build RESTful APIs for secure data exchange between internal banking systems, improving data accessibility and integration."
- Implemented Java based data validation checks within the ETL processes, improving data quality within financial reporting.
- Contributed to the development of Java-based data processing jobs for optimizing SQL query performance, reducing processing times for critical financial reports.
- Designed and implemented Java servlets for handling secure online transactions, ensuring compliance with financial industry standards.

- Utilized Java Database Connectivity (JDBC) to develop data access layers for efficient interaction with the bank's core database systems.
- Collaborated with business teams and senior analysts to define key performance indicators - KPIs and develop custom financial reports for executive decision-making.
- Performed ad-hoc financial analysis, helping leadership teams optimize loan approvals, reduce non-performing assets (NPAs), and improve credit risk assessment.
- Optimized SQL query performance by indexing and tuning database structures, reducing query execution time by 30% for critical banking reports.
- Worked closely with cross-functional teams like risk, finance, and IT to bridge the gap between business needs and data-driven solutions.

**Environment:** Databases: SQL Server, Oracle, PostgreSQL | Data Visualization: Power BI, Tableau, Excel | Programming & Scripting: SQL, Python, Pandas, NumPy, Matplotlib, Seaborn | Analytics & Reporting: SQL Reporting Services, Advanced Excel-Pivot Tables, Macros, Statistical Modeling | Data Processing & ETL: SQL Stored Procedures, Functions, Data Cleaning, Query Optimization | Finance-Specific Tools: Risk & Compliance Reporting, Loan Performance Analysis, Fraud Detection Analytics | JavaScript, Java database connectivity

### **CERTIFICATION**

- **IBM** Certified DATA SCIENCE TOOLS
- **IBM** Certified R for Data Science
- **AWS** Certified Data Engineer

### **EDUCATION**

**UMassd, MA, USA**

Data Science (*GPA: 3.5/4.0*)

Aug 2021 - May 2023

**Godavari institute of Eng & Tech, A.P, India**

Computer Science & Engineering (*CGPA: 8.5/10.0*)

March 2013 - May 2017